

Module Development for a High Specific Power Density High-Efficiency Cryogenic Solid-State Circuit Breaker for Electrified Aircraft Propulsion

Shimul K. Dam¹, Member, IEEE, Ching-Hsiang Yang², Student Member, IEEE,
Zhou Dong³, Senior Member, IEEE, Dehao Qin⁴, Student Member, IEEE,
Ruirui Chen⁵, Member, IEEE, Fred Wang⁶, Fellow, IEEE,
Hua Bai⁷, Senior Member, IEEE, and Zheyu Zhang⁸, Senior Member, IEEE

Abstract—Solid-state circuit breaker (SSCB) with high efficiency and high gravimetric power density is required for electrified aircraft propulsion (EAP). A cryogenically cooled gallium nitride (GaN) device-based SSCB module is developed for the EAP system to achieve a higher power density than existing SSCBs. The SSCB module is suitable for medium voltage direct current (MVdc) applications and is designed with low-weight modular cold plates. Each module is rated for 750 V, 100 A, and can interrupt a fault current of 1 kA at cryogenic temperature. The SSCB module structure design, thermal interface design, modular cold plate design, and insulation design are discussed. The developed SSCB module prototype achieves a power density of 329 kW/kg and an efficiency of 99.93%. Experimental results of interrupting a fault of 1 kA and insulation test of the SSCB module are provided to validate its operation.

Index Terms—Aircraft propulsion, cryogenic cooling, GaN HEMT, solid-state circuit breaker (SSCB).

I. INTRODUCTION

THE power distribution system of a fully electric aircraft propulsion (EAP) system is expected to use medium

voltage dc (MVdc) to transfer mega-watts of power [1]. Such a system has fast-rising fault current due to medium voltage and low line inductance. Moreover, the EAP system is dominated by power converters with low surge capability [2], reinforcing the need for very fast fault current interruption by a circuit breaker. The circuit breaker should also be lightweight and avoid any nuisance trip by allowing normal transient currents. Thus, the circuit breaker in this application should also have a high breaking capacity along with a high specific power density [3].

A traditional mechanical circuit breaker (MCB) [4] is heavy, high-maintenance, and requires tens of milliseconds to interrupt a fault current. A hybrid circuit breaker (HCB) [5] improves the interrupt time to a few milliseconds but still is bulky and too slow to interrupt a fast-rising fault current in an MVdc system [3]. A solid-state circuit breaker (SSCB) can achieve both fast fault interruption and low weight compared to MCB and HCB [6]. However, the efficiency and breaking capacity of the SSCB are lower than its counterparts due to the use of semiconductor switches in the conduction path [7].

Thyristors are frequently used as switching devices in SSCB due to their high current and voltage ratings, high breaking capacity, and low cost [8], [9], [10], [11]. However, the forced current commutation to turn off thyristors leads to additional delay in response and bulky design with large passive elements for commutation circuits. Si IGBTs are utilized to achieve faster fault interruption and simple drive circuits at the expense of higher conduction loss [12] compared to thyristor-based solutions. In order to leverage the superior material properties of wide band-gap (WBG) devices, the SiC and gallium nitride (GaN) devices have been utilized in SSCB applications [13], [14], [15], [16], [17]. However, even with the superior performance of WBG devices, the SSCBs still have lower efficiency and breaking capacity compared to MCBs and HCBs [18].

A way to improve the efficiency and breaking capacity of WBG-based SSCBs is to parallel several devices. However, paralleling more devices leads to increased dynamic current unbalance and instability issues during high fault current interruption [19], [20]. Moreover, a large number of devices increases cost and reduces specific power density due to increased PCB and enclosure size.

Manuscript received 30 November 2023; revised 31 March 2024 and 7 June 2024; accepted 29 June 2024. Date of publication 9 July 2024; date of current version 4 September 2024. This work was primarily supported by ARPA-e, Department of Energy, USA, under the award number DE-AR0001467. This work made use of Engineering Research Center Shared Facilities supported by the Engineering Research Center Program of the National Science Foundation and the Department of Energy under NSF Award Number EEC-1041877 and the CURENT Industry Partnership Program. Recommended for publication by Associate Editor M. Saeedifard. (Corresponding author: Shimul K. Dam.)

Shimul K. Dam, Ching-Hsiang Yang, Ruirui Chen, and Hua Bai are with the Department of Electrical Engineering and Computer Science, University of Tennessee, Knoxville, TN 37916 USA (e-mail: sdam@utk.edu; cyang30@vols.utk.edu; rchen14@vols.utk.edu; hbai2@utk.edu).

Zhou Dong is with ABB US Research Center, ABB, Inc., Raleigh, NC 27606 USA (e-mail: zhou.dong@us.abb.com).

Dehao Qin is with the College of Engineering, Computing and Applied Sciences, Clemson University, North Charleston, SC 29405 USA (e-mail: dehaoq@g.clemson.edu).

Fred Wang is with the Department of Electrical Engineering and Computer Science, University of Tennessee, Knoxville, TN 37916 USA, and also with the Electrification and Energy Infrastructures Division, Oak Ridge National Laboratory, Oak Ridge, TN 37830 USA (e-mail: fred.wang@utk.edu).

Zheyu Zhang is with the Department of Electrical, Computer and Systems Engineering, Rensselaer Polytechnic Institute, Troy, NY 12180 USA (e-mail: zhangz49@rpi.edu).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TPEL.2024.3425586>.

Digital Object Identifier 10.1109/TPEL.2024.3425586

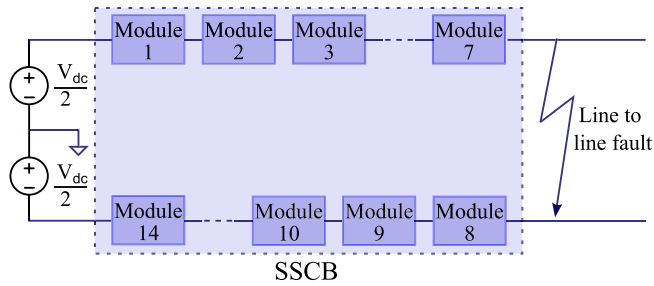


Fig. 1. SSCB in a bipolar DC distribution system under line-to-line fault condition.

The efficiency, breaking capacity, and specific power density of an SSCB can be significantly improved by using cryogenic coolant available in an EAP system that uses liquid hydrogen or liquified natural gas as fuel or high-temperature superconducting (HTS) machines for propulsion [21]. Studies of characteristics of semiconductor devices in literature [22], [23] have revealed that their conduction loss can be reduced at low temperatures. This effect is found to be most pronounced in GaN devices, which show a monotonic decrease of ON-state resistance $R_{ds(on)}$ with the reduction of temperature down to 4 K [24]. The $R_{ds(on)}$ for some GaN devices can be reduced as much as five times at liquid nitrogen (LN_2) temperature [22], [23]. Also, due to cryogenic junction temperature ($< -150^\circ\text{C}$) operation at rated current, the GaN device can withstand higher than rated current for a longer duration before its junction temperature reaches the maximum allowable temperature of 150°C . Thus, the efficiency and breaking capacity target of the SSCB can be met with fewer paralleled devices, minimizing the drawbacks associated with device paralleling.

Cryogenic cooling has been demonstrated to implement an HTS conductor for current limiting [25], [26] and to improve the efficiency of the Si-based 24-V 85-A SSCB [25]. However, HTS conductors can take 0.5–1 ms to respond and the use of inductors and capacitors can reduce the power density of the SSCB in [26]. Also, the breakdown voltage of Si devices reduces at low temperatures [22], unlike the GaN devices, causing underutilization of the device voltage rating. Apart from these works, the advantages of cryogenically cooled SSCB are hardly explored in existing literature.

The specific power density of the circuit breaker in an all-electric aircraft to outperform the conventional aircraft needs to be high [3]. In literature, the work in [17] reported the highest specific power density of 112 kW/kg with an efficiency of 99.5%. The SiC power modules, cold plate, and enclosure are together responsible for 60% weight in [17]. Thus, improved semiconductor performance, lightweight thermal design, and compact mechanical and insulation design can help to greatly improve the specific power density. This work presents the development of a cryogenically cooled 750-V, 100-A SSCB module for bipolar MVdc applications up to 10 kV. According to the standard DO-160G [27] for airborne equipment, the SSCB may be subjected up to 120% of rated voltage as an abnormal steady-state voltage. The module is designed to withstand 120% of rated voltage and is capable of interrupting a fault current of 1 kA ($10\times$ rated current). For a 10-kV bipolar MVdc system, 14 such modules

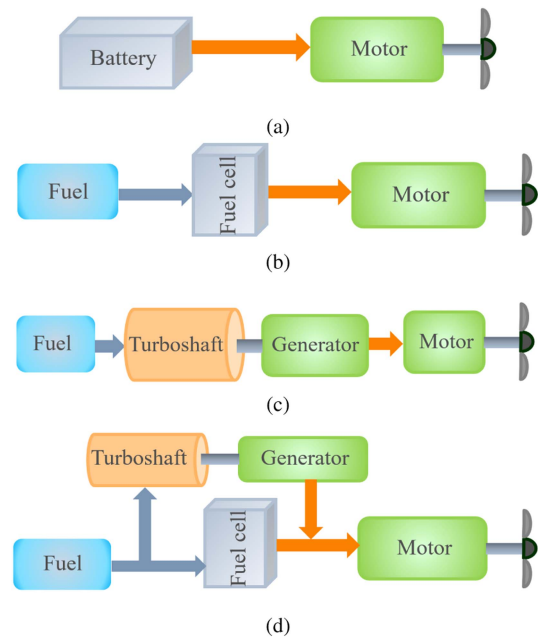


Fig. 2. Electric aircraft propulsion configurations. (a) Fully electric. (b) Fuel-cell. (c) Turbo-electric. (d) Turbo-electric with fuel cell [1].

are needed and they are symmetrically placed on positive and negative lines to reduce voltage insulation requirement to 5 kV, as shown in Fig. 1. Considering 120% of rated voltage, each module insulation is designed to withstand 6 kV. This work achieves high specific power density by using the following.

- 1) Only two GaN dies in parallel for compact design and low cost while achieving high efficiency with the aid of cryogenic cooling.
- 2) Thin-walled light-weight aluminum cold plate design with suitable cryogenic sealing.
- 3) GaN dies with appropriate cryogenic thermal interface design instead of bulky power modules to improve specific power density.
- 4) Cryogenic temperature compatible coating based insulation design to achieve compact SSCB design.

This article is an enhanced version of the preliminary work presented in [28]. The additional contents include a detailed literature review to highlight the contributions, detailed design discussion and mechanical stress simulation verification of cold plate design, thermal interface design for cryogenic thermal cycling, detailed discussion on insulation requirement, cold plate cooling-based test results at different fault currents, and efficiency measurement results.

The rest of this article is organized as follows. Section II describes the cryogenic cooling system. The design of an SSCB module including mechanical design, cold plate design, and thermal interface design is discussed in Section III. Insulation design and electric field simulation are presented in Section IV, and the experimental results are presented in Section V. Finally, Section VI concludes this article.

II. CRYOGENIC COOLING SYSTEM

An electrified aircraft propulsion system uses electric motors for propulsion. The required electric power may be generated

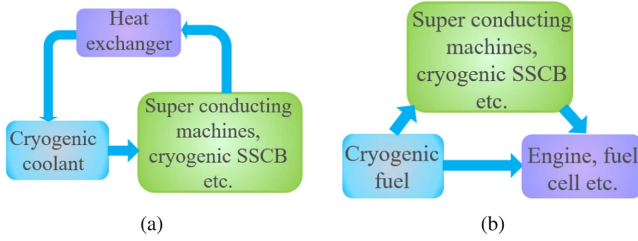


Fig. 3. Cryogenic cooling configurations. (a) Closed-loop cooling. (b) Open-loop cooling.

from different sources, such as batteries, fuel cells, and turboshaft generators [1]. Thus an EAP can have different configurations, as shown in Fig. 2. A fully electric EAP uses a battery bank to generate power for propulsion. However, the energy density of battery technology will not be sufficient for commercial single-isle or double-isle aircraft in the near future [29]. An alternate approach is the use of fuel cells to improve energy density. However, due to the limited power ratings of the fuel cells, a more practical approach for commercial aircraft EAP is to use a turbo-shaft engine to drive a generator and produce electric power, as shown in Fig. 2(c). A combination of turbo-shaft and fuel cell is also possible, as shown in Fig. 2(d). Apart from the battery-based configuration, all other EAPs use either cryogenic fuels, such as liquid hydrogen (-253°C) and liquified natural gas (-162°C) or noncryogenic fuel, such as ammonia (-33°C).

The generators and motors in future electrified propulsion systems are expected to have cryogenic cooling for low-weight compact design [29]. If no cryogenic fuel is available, a cryogenic cooling system is necessary that works in closed loop, as shown in Fig. 3(a). The same coolant can be utilized for SSCB cooling. This process raises the temperature of the coolant. Thus, the coolant needs to be refrigerated before it returns to the coolant storage.

However, if cryogenic fuel is available, then the need for a closed-loop cooling system can be avoided. In an open-loop cooling system, the cryogenic fuel is used as coolant which is fed to the turbo engine or fuel cell, as shown in Fig. 3(b). The engine or fuel cell does not need fuel to be at cryogenic temperature. In fact, fuel needs to be heated up for their proper operation [30]. As the cryogenic fuel does not return to its cryogenic storage after cooling operation, it does not need any refrigeration. The heat from electronics and machines helps to raise fuel temperature for proper operation of the engine and fuel cell.

III. SSCB MODULE DESIGN

The SSCB module has four 650-V/150-A GaN dies [31] as switching devices. Each GaN die is paralleled with an R - C snubber and a transient voltage suppressor (TVS) diode. The R - C snubber helps to reduce dV/dt and loss during the turn-OFF transition under a large fault current. The TVS diode absorbs the fault energy and extinguishes the fault current quickly. The module is designed to carry 100-A rated current with over 99.9% efficiency and to interrupt a maximum of 1-kA fault current at cryogenic temperature ($<-150^\circ\text{C}$).

Apart from the switching devices, each module has other important electrical components and circuits, such as gate drivers,

signal isolators, control logic circuits, fault detection circuits, and isolated power supplies. Each isolated power supply can withstand a voltage higher than the maximum voltage between any module and the cold plate. So, the primary side terminals of all the isolated power supplies are connected together to a single supply, whose ground is connected to the cold plates. To achieve a compact design, the module components are placed on two separate boards: 1) the main board and 2) the auxiliary board. The main board has the GaN dies, snubber circuit, TVS diode, and gate driver. The auxiliary board contains all other components, which detect faults, receive control commands from the controller, and communicate with other modules. This structure of the module is shown in Fig. 4.

While low temperature is more beneficial for efficiency improvement due to monotonic reduction in $R_{ds(on)}$ of GaN device, all other components of the SSCB module do not have any advantage at low temperature. In fact, these components are usually rated for temperatures higher than -40°C and need to be characterized at cryogenic temperature to verify their performance. The designed SSCB module uses LN_2 for cooling, which has a temperature of -196°C . However, due to power loss in dies and heat leakage from the external environment, the module is expected to operate around -180°C . So, the components selected for this work have been characterized down to -180°C [32]. Many of the commercially available components do not perform well at cryogenic temperatures, leading to additional constraints on the SSCB design. Some of these constraints are as follows.

- 1) Several commonly used capacitors perform poorly compared to C0G capacitors. However, C0G capacitors are costly and not available with high capacitance ratings.
- 2) Many ICs may need to be replaced with alternative ICs, which may increase cost.
- 3) Fiber optics are excellent means for medium voltage signal isolation, but do not work well at cryogenic temperature.

A. Operating Principles

If the SSCB has n modules in series to withstand a system voltage of V_{dc} , then each module has to withstand a voltage

$$V_{dc_mod} = \frac{V_{dc}}{n}. \quad (1)$$

The rated module current is the same as the rated SSCB current. Since the stored energy in the line inductor is proportional to its inductance, the energy to be absorbed by a single module is the energy stored in an equivalent inductance given as follows:

$$L_{line_mod} = \frac{L_{line_mod}}{n}. \quad (2)$$

Using (1) and (2), the equivalent circuit for a single module under short-circuit fault condition is given in Fig. 5(a). The SSCB module carries a load current i_{load} which is shared by two parallel devices. The voltage and current waveforms of a module during a line-line fault are shown in Fig. 5(b). The waveforms can be divided into five zones. Zone 2 and 3 have short durations and are shown in the zoomed transition waveforms on the right side of Fig. 5(b).

The zones of operation are described as follows.

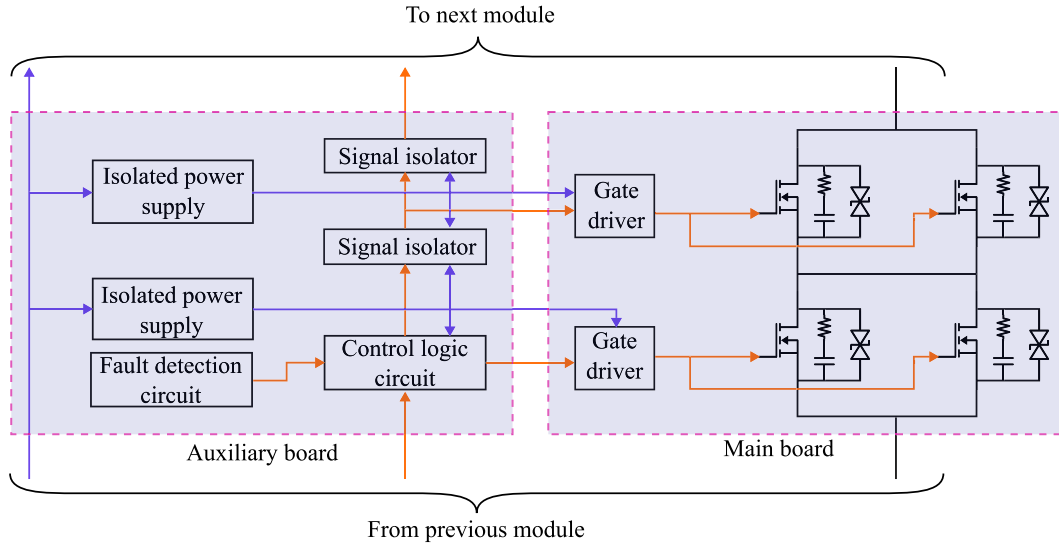


Fig. 4. Architecture of a module of the proposed modular SSCB.

- 1) *Zone 1*: Zone 1 starts when a line-to-line fault occurs in Fig. 5(a), and the current through the SSCB module increases rapidly. When the module current is higher than the protection threshold, the switches are turned OFF and the module enters zone 2.
- 2) *Zone 2*: At the beginning of this zone, current commutes from switch to snubber. As a result, the snubber capacitor is charged, and drain-source voltage V_{ds} increases. The fault current still increases in this zone as V_{ds} is lower than V_{dc_mod} . The module enters Zone 3 when V_{ds} exceeds V_{dc_mod} .
- 3) *Zone 3*: This zone continues until the V_{ds} reaches peak clamping voltage of TVS. Since $V_{ds} > V_{dc_mod}$, the fault current starts to fall in this zone.
- 4) *Zone 4*: At the beginning of this zone, the V_{ds} falls from peak clamping voltage to average clamping voltage following the TVS characteristics. V_{ds} remains at the average clamping voltage of TVS until the fault current falls to zero. The TVS then stops conducting and the module enters zone 5.
- 5) *Zone 5*: As the TVS stops conducting, the voltage V_{ds} falls from the average clamping voltage to V_{dc_mod} . This change in voltage across the snubber capacitor leads to an LC resonance between line inductor L_{line_mod} and snubber capacitor C_{snb} . The LC resonance is observed in V_{ds} , module current i_{load} , and snubber current i_{snb} with following voltage peak, current peak, and resonant frequency,

$$V_{ds}^{pk} = (V_{cl_avg}) \quad (3)$$

$$i_{snb}^{pk} = \left(V_{cl_avg} - \frac{V_{dc_mod}}{2} \right) \sqrt{\frac{C_{snb}}{L_{line_mod}}} \quad (4)$$

$$f_r = \frac{1}{2\pi \sqrt{L_{line_mod} C_{snb}}} \quad (5)$$

B. Selection of TVS Diode

Two popular energy absorption devices are metal oxide varistors (MOVs) and TVS diodes. Compared to TVS diodes, the MOVs have a shorter life, slower response, and higher clamping factor (ratio of peak and average clamping voltages) [33], [34], [35]. Thus, when operated within its ratings, the TVS diodes offer higher reliability [36] and lower voltage stress on switching devices, enabling higher utilization of the voltage rating of switches. However, TVS diodes are more expensive and are available at lower power and voltage levels than the MOVs, limiting their applications.

In the proposed work, the SSCB is modular, where each module is rated for 750 V and uses two 650-V GaN dies in series. At this voltage range, it is possible to achieve enough average clamping voltage for each GaN die using two TVS diodes in series. Thus, TVS diodes can satisfy the requirement in this work and provide a longer life span, lower stress on GaN dies, and faster response as well.

The TVS ratings must be selected to satisfy the maximum allowable current interruption time t_{int} for the peak load current i_{load}^{pk} . These requirements are specified based on system fault study and the overcurrent withstand capabilities of system components. In Zone 4 of Fig. 5(b), each TVS has to carry half of the load current, and the slope of the current is given by $(2V_{cl_avg} - V_{dc_mod})/L_{line_mod}$. Thus, the required average clamping voltage can be obtained as follows:

$$V_{cl_avg} \geq \frac{V_{dc_mod}}{2} + \frac{i_{load}^{pk} L_{line_mod}}{2t_{int}} \quad (6)$$

In the worst case, the energy absorbed by the TVS during Zone 4 is given by

$$E_{tvs} = \frac{i_{load}^{pk} V_{cl_avg}}{4} t_{int} \quad (7)$$

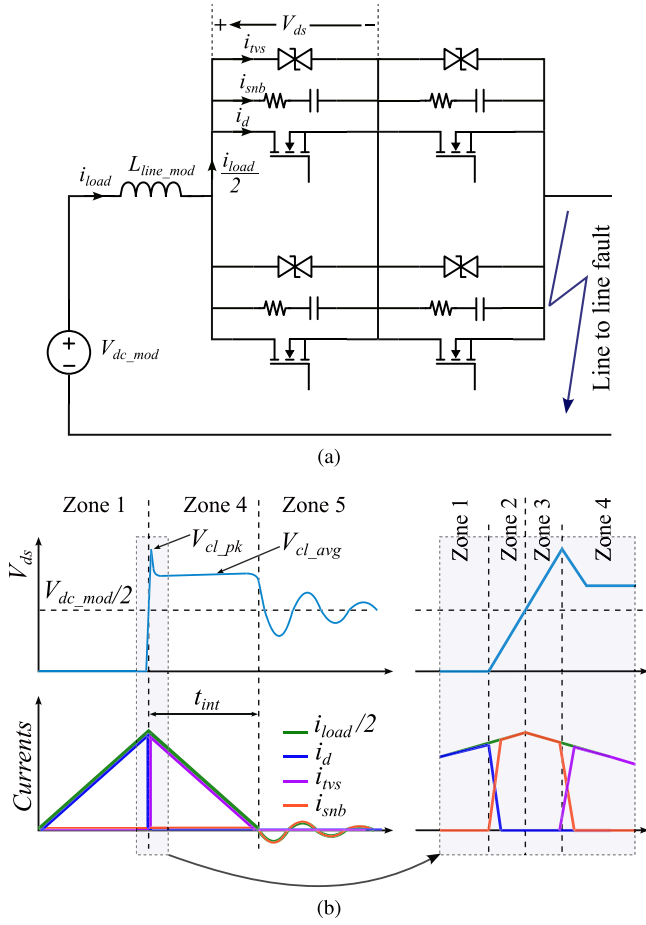


Fig. 5. (a) SSCB module under line-line sort-circuit fault condition. (b) Voltage and current waveforms during fault interruption.

Thus, TVS diode is selected to satisfy (6) based on the average clamping voltage and the energy absorption requirement in (7) based on the peak pulse power rating in the datasheet. In this work, each TVS in Fig. 5(a) is implemented with a series combination of two TVS diodes AK3-430 C and AK3-150 C from Littelfuse [37] to meet both criteria. The average clamping voltage requirement is verified using the cryogenic characterization of these diodes in [38]. Since there are two sets of paralleled TVS diodes in one SSCB module, the energy absorption rating of the module is given by

$$E_{\text{mod}} = i_{\text{load}}^{\text{pk}} V_{\text{cl-avg}} t_{\text{int}}. \quad (8)$$

The selected TVS diodes from Littelfuse have been characterized in [38] at cryogenic temperature. It is observed that the peak clamping voltage of the TVS diode reduces at lower temperatures, leading to lower voltage stress on the GaN die. Hence, at lower temperatures, the TVS diode can be used to interrupt a higher current compared to room temperature without exceeding allowable voltage stress on the die.

A more detailed discussion on the electrical design of different components of the SSCB module is reported in [38]. The following sections discuss the mechanical, thermal interface, and insulation design of the SSCB module to achieve high specific power density at medium voltage with cryogenic cooling.

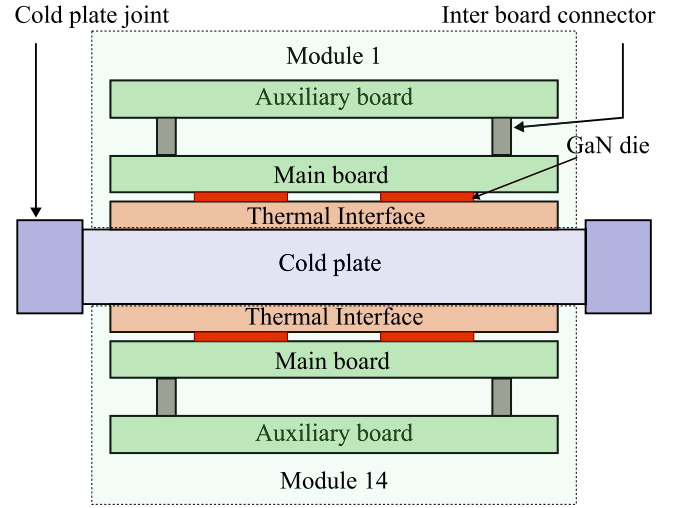


Fig. 6. Mechanical structure of two SSCB modules and one cold plate assembly.

C. Mechanical Structure

The mechanical structure of the module and cold plate assembly is shown in Fig. 6. Two SSCB modules are attached to one cold plate to improve the specific power density and reduce cost. For each module, the gate, drain, and source pads on the top surface of the 650-V, 150-A GaN die [31] are soldered on the bottom side of the main board PCB. The bottom surface of the die is the silicon substrate, which provides the main thermal conduction path from the die junction to the cold plate. A thermal interface is designed to interface between the die substrate and the cold plate and is designed to withstand mechanical stress during cryogenic thermal cycling.

The cryogenic SSCB is cooled using LN_2 in this work. In this design, the main boards with GaN dies are closer to the cold plate and the auxiliary boards are away from the cold plate. This arrangement provides some thermal insulation to the auxiliary board components, which do not need to be at cryogenic temperature. Interboard signal connectors are used to send the signals from the auxiliary board to the main board and to receive the measurement signals. The main and auxiliary boards are mechanically supported by nylon spacers and screws. The spacer length between the auxiliary and main board is 20 mm to provide enough clearance for main board components from the auxiliary board PCB. Large spacing between two boards is better for thermal isolation but leads to a less compact design and increased enclosure size and weight.

D. Cold Plate Design

The cold plate design for liquid cooling usually uses many fins in a cold plate channel or many microchannels in a solid aluminum block [39], [40] to increase surface area. Since the LN_2 has good thermal conductivity [41], the cold plate surface area requirement is low. Thus, a lightweight cold plate can be designed to carry the LN_2 by simply using a hollow aluminum tube. A thin-walled rectangular aluminium tube is selected as a cold plate to reduce weight further. However, the use of rectangular thin-walled aluminum tubes leads to two major

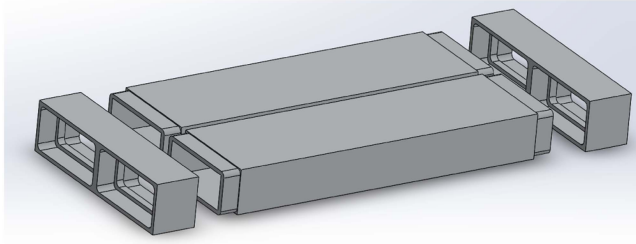
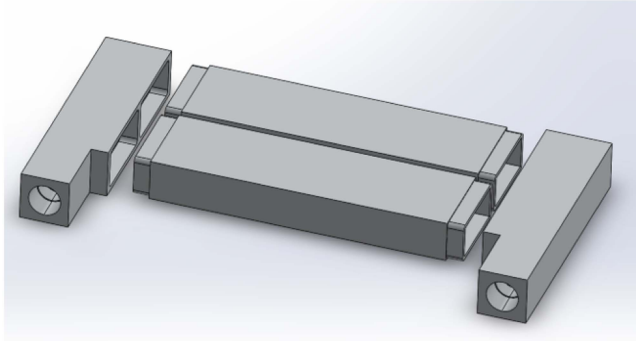


Fig. 7. Modular cold plates and cold plate interface.

Fig. 8. LN₂ tube to cold plate interface.

challenges. First, due to the rectangular shape of the tubes, the usual round-shaped tube fittings cannot be used to join two cold plates during the series connection of modules. Second, the sealing of the joints is very challenging due to the thin walls and cryogenic temperature.

Manufacturing a custom-made rectangular tube with a thin wall can be expensive and time-consuming. Hence, a commercially available cold plate with a thin wall (1.6 mm) is selected and two such tubes are connected in parallel to cover all the dies of the module. The modular design of the cold plates is achieved by designing an additional part as the joint between the cold plates of two modules. The designed cold plates and the joints are shown in Fig. 7. It should be noted that the parts of the cold plate surface near the ends are machined away so that they can fit into the slots of the joint parts.

The joints can connect two sets of cold plates. However, it cannot directly interface with an LN₂ tube. Thus, to interface with the external cryogenic cooling system, a tube to cold plate interface is designed, as shown in Fig. 8. Two such interface parts are needed to connect the terminals of the cold plate assembly of one or more SSCB modules to the LN₂ carrying tubes. Using these designs, multiple parts can be joined in a modular way. However, reliable sealing of the joints at cryogenic temperature is still a challenge.

1) *Joining Method for Cold Plate Parts:* Ensuring good sealing when joining two separate parts is difficult at cryogenic temperatures due to the coefficient of thermal expansion (CTE) mismatch. The sealing material is subjected to large mechanical stress during cryogenic thermal cycling. Commonly used sealing material, such as epoxy or rubber lose their flexibility at cryogenic temperatures and are likely to crack. An alternate approach

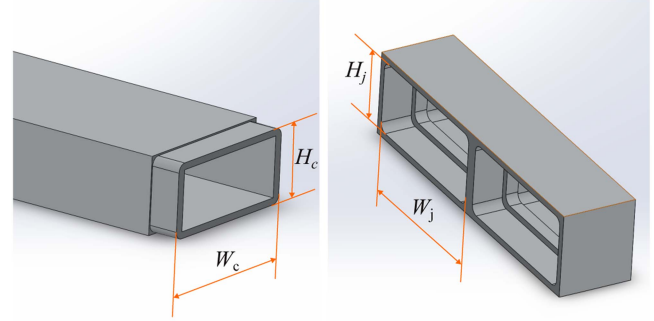


Fig. 9. Mating parts of cold plate and joint slots.

is to weld the parts together to avoid using another material and CTE mismatch. However, welding thin-walled aluminum parts is very challenging due to the low melting point of aluminum and may damage the parts in the process. A more reliable heat-shrink method is selected in this work to join the cold plates with the joints.

In the heat shrink method, the cold plate and the joint are designed in such a way that the height H_c and width W_c of the mating part of the cold plate in Fig. 9 are slightly higher than the height H_j and width W_j , respectively, of the joint slot. This difference is called interference and is defined as

$$\text{Interference along height, } H_i = H_c - H_j \quad (9)$$

$$\text{Interference along width, } W_i = W_c - W_j. \quad (10)$$

During the assembly process, the cold plate is cooled down using dry ice and the joint is heated with a hot plate to a sufficiently high temperature that ensures $H_c < H_j$ and $W_c < W_j$. Under such conditions, the cold plate can be inserted into the joint slots. The cooling and heating sources are then removed, causing the expansion of the cold plate and shrinking of joint slots to create a strong metal seal. Since no other material is involved here and no CTE mismatch is present, the seal can withstand large thermal cycling. The joint surfaces do not need very high-precision finishing. A standard mechanical polishing of the surfaces is sufficient.

2) *Interference Design for Heat Shrink Method:* While designing the interference, the mechanical stresses on the heat shrink joints need to be considered. The stresses on the cold plate mating parts are given as

$$\text{Stress along height, } \sigma_h = E \frac{H_i}{H_c} \quad (11)$$

$$\text{Stress along width, } \sigma_w = E \frac{W_i}{W_c} \quad (12)$$

where E is the Young's modulus of Aluminum. Stress on the joint slots can also be evaluated with these expressions by replacing H_c and W_c with H_j and W_j . However, since the interference is a very small fraction of height and width, $H_c \simeq H_j$ and $W_c \simeq W_j$, leading to almost equal stress on the joint slots. For a successful metal seal, the following conditions must be met:

$$\max [\sigma_h, \sigma_w] < \sigma_{AL} \quad (13)$$

TABLE I
MAXIMUM AND MINIMUM STRESSES IN HEAT SHRINK-BASED JOINTS

	H_c (mm)	W_c (mm)	H_i (mil)	W_i (mil)	σ_H (psi)	σ_W (psi)
Maximum interference	11.66	24.92	1.0	1.0	21768	10185
Minimum interference	11.66	24.92	0.2	0.2	4354	2037

$$\min[\sigma_h, \sigma_w] > P_{LN2} \quad (14)$$

where σ_{AL} is the yield strength of aluminum, which is 40 000 psi for selected aluminum alloy 6061 [42] and P_{LN2} is the pressure of LN_2 , which in this work is 50 psi, preset by the supplier.

The interference value also depends on manufacturing tolerance, which leads to a range for the interference, leading to a range for the resulting stresses. The inequalities in (13) and (14) must be valid for maximum stress and minimum stress conditions, respectively. Although a tight manufacturing tolerance is desired, such a tolerance requires increased machining efforts and, in turn, increased cost. A manufacturing tolerance of ± 0.2 mil is considered for each dimension in this work based on the manufacturer's capability. The interference is chosen as 0.6 mil along both height and width. Considering ± 0.2 -mil tolerance, the interference can be in the range of 0.2–1.0 mil.

The stresses for the minimum and maximum interferences are calculated in Table I. It can be observed that the maximum stresses are below the yield stress of 40 000 psi and the minimum stresses are higher than the LN_2 pressure of 50 psi. Thus, both the conditions in (13) and (14) are satisfied. However, It should be noted that the stresses on the corners of the cold plate and joint mating parts are higher than stresses in other areas due to stress from both directions. A stress simulation is performed in the next section to ensure that the peak stress in the corners is still within the maximum allowable limit.

3) *Mechanical Stress Simulation*: Mechanical stress simulation is performed in COMSOL to verify the interference design. The simulation results for the nominal interference of 0.6 mils in Fig. 10(a) show that the peak stress on both cold plate and the joint is below the yield strength of aluminum, which is 40 000 psi. On the other hand, for the maximum interference of 1 mil, the peak stress comes close to the yield strength locally, as shown in Fig. 10(b) in the corners, which is not enough to damage the aluminum wall. An interference higher than 0.6 mils can lead to exceeding yield strength across the wall thickness and cause seal failure.

E. Thermal Interface Design

The thermal interface between the die and the cold plate has to provide a good thermal conduction path as well as good electrical insulation. Commonly used soft thermal pad materials, such as silicone rubber become stiff at cryogenic temperature and crack under thermal cycling. Although these materials still can work as good thermal conductors, they cannot provide electrical insulation due to cracks. Instead, ceramic-based direct bonded copper (DBC) is selected in this work. Aluminum nitride

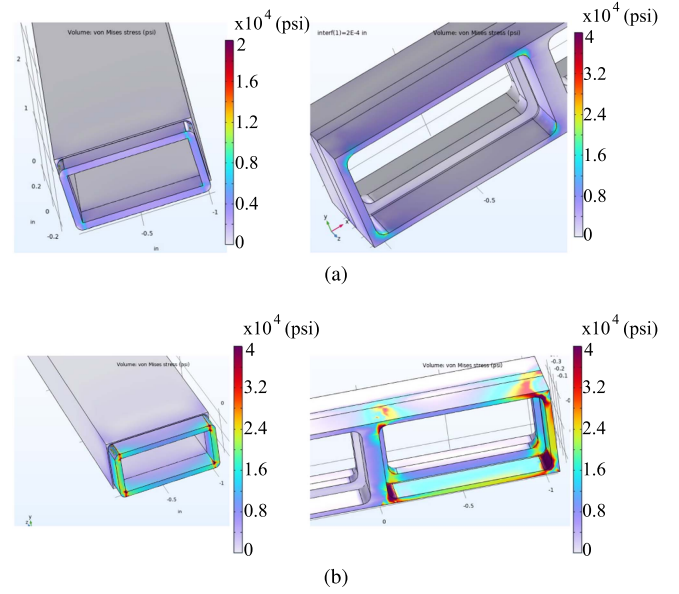


Fig. 10. Simulated mechanical stress on heat-shrink joints at 77 K for (a) 0.6-mil interference and (b) 1-mil interference.

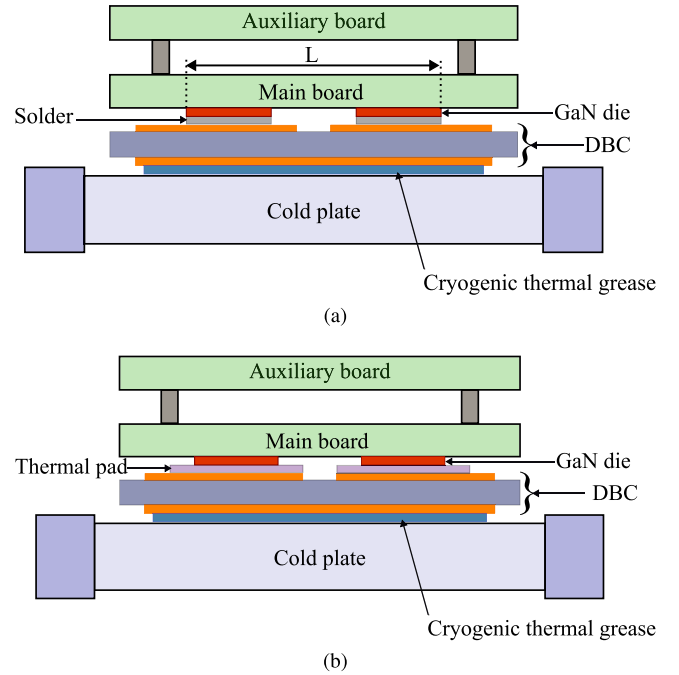


Fig. 11. Die to cold plate interface through DBC using (a) solder and (b) thermal pad.

ceramic-based DBC is chosen with both side copper layers, which help to provide mechanical strength to the DBC structure.

The simplest way to implement the thermal interface using DBC is to solder the die substrate on the top layer of DBC, as shown in Fig. 11(a). The DBC is then placed on the cold plate. A cryogenic thermal grease is applied between the DBC bottom surface and the cold plate to avoid any air gap. However, due to different CTE, there is a mismatch of thermal expansion between

PCB and DBC, as given by

$$\Delta L = L(\alpha_{FR4} - \alpha_{DBC})\Delta T \quad (15)$$

where L is the maximum distance between the pads of two dies, as shown in Fig. 11(a), α_{FR4} and α_{DBC} are the CTE of FR4 and DBC, respectively, and ΔT is the maximum change in temperature during thermal cycling. The CTE of FR4 material is 12–14 ppm/°C [43] and that of DBC is 4.7 ppm/°C [44]. The resulting stress from this CTE mismatch on the die pads is

$$\sigma_{PCB} = E_{FR4} \frac{\Delta L}{L} \quad (16)$$

where E_{FR4} is the Young's modulus of FR4 material, which is 3.5×10^6 psi [43]. In this work, L is 35 mm and ΔT is 200 °C, which results in a stress of 6510 psi on the die. As a result, the die is bent and eventually breaks under high stress. Since the maximum safe stress on the die is not provided by the manufacturer, it is experimentally verified that the die cannot take this much stress and cracks under thermal cycling.

Hence, a modified structure is designed here to relieve the mechanical stress. In the new structure in Fig. 11(b), the dies are not soldered to the DBC. A layer of thermal pad is placed between the die and DBC. In this structure, a thermal pad only helps to provide a softer interface to the die. Even if the thermal pad cracks, the DBC can provide sufficient electrical isolation. Since the thermal pad and DBC are both good thermal conductors, this thermal interface structure can provide good thermal conduction, electrical isolation, and a softer interface for the die. During thermal cycling, the thermal pad allows the die to move laterally as the PCB expands and shrinks, relieving additional mechanical stress due to CTE mismatch. This structure too is tested experimentally and it is found that the die can survive the thermal cycling.

F. Thermal Performance Simulation

The worst-case temperature rise in dies occurs when the high fault current flows through the module. The SSCB module can withstand an overcurrent condition for a limited duration. This withstand time is lowest for the peak fault current of 1 kA. A thermal simulation is performed on the GaN dies with liquid Nitrogen cooling to determine the overcurrent withstand duration of the module. The simulation result in Fig. 12 shows that the module can carry 1 kA for 56 ms without exceeding the junction temperature limit of 150 °C. The controller and the desat protection of the module are programmed to open the breaker before the overcurrent duration exceeds its allowable limit.

IV. INSULATION DESIGN OF THE MODULE

Due to the compact design of the module, the voltage isolation requirement between two close layers can be high. Hence, the insulation should be carefully designed.

A. Insulation Requirement

As shown in Fig. 1, the 10-kV SSCB is designed for a bipolar MVdc system. The midpoint of the dc link is grounded.

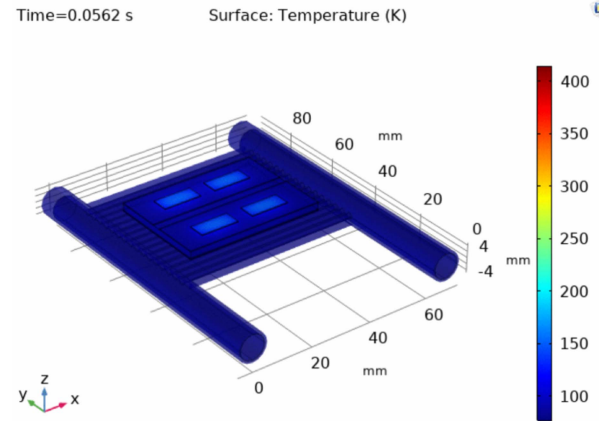


Fig. 12. Simulation of SSCB module temperature rise under 1-kA fault current condition after 56 ms.

Out of 14 series connected SSCB modules, modules 1 to 7 are connected to the positive bus, and modules 8 to 14 are connected to the negative bus. Such a split arrangement helps to reduce the insulation requirements of the SSCB module. In case of a line–line fault, all the modules come in series to withstand the full 10-kV voltage.

The cold plates of the SSCB are connected to an external cooling system with metal tubes. Hence, the cold plates are kept grounded for safety reasons. One SSCB module from the positive dc bus is placed on the top side of a cold plate and one module from the negative dc bus is placed on the bottom side of the cold plate. For example, modules 1 and 14 of Fig. 1 are placed on one cold plate, as shown in Fig. 6. Thus, the voltage isolation between each module and cold plate is 5 kV. However, a 120% abnormal steady-state voltage is considered in this work, which increases the isolation requirement to 6 kV.

The rated voltage for each module is 750 V which is different from the insulation voltage of the module. When the SSCB is open, each module blocks up to 750 V, which is the voltage difference between two adjacent modules. The insulation between the cold plate and each module has to withstand a voltage of up to 6 kV when the SSCB switches are turned ON. Allowing some more safety margin, the insulation is designed here considering an isolation requirement of 8 kV.

B. Insulation Design

The DBC layer in the thermal interface between the die and cold plate provides the electrical isolation. The DBC uses aluminum nitride (AlN) as ceramic material, which has an electrical breakdown strength of 20 kV/mm [44]. A standard thickness of 0.64 mm is selected for the AlN layer which can withstand up to 12.8 kV. Since the copper layer on the DBC does not carry any current, its thickness can be reduced to improve module weight. A copper layer of 0.1-mm thickness is used here for the DBC design, which is the lowest thickness the manufacturer offers. The designed and manufactured DBC is shown in Fig. 13. The top copper surface is split into two to provide isolation between the upper and lower dies in the module.

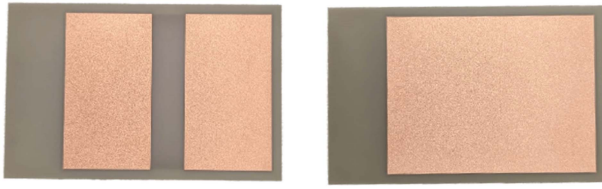


Fig. 13. Top side and bottom side of DBC.

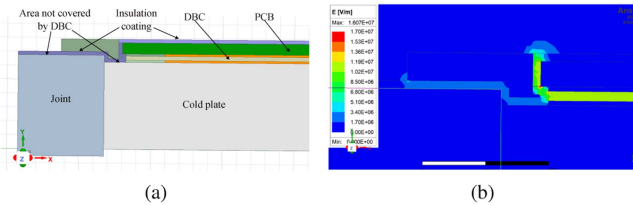


Fig. 14. Electrostatic simulation model for cold plate-DBC-PCB assembly and simulated electric field.

It can be observed from Fig. 14(a) that a part of the cold plate and the joints are not covered by the DBC. Hence, there can be electrical breakdown, such as flashover from module PCB or DBC top surface to these exposed cold plate areas. To avoid this breakdown, a layer of insulating coating is applied on the exposed area. The insulation coating material selected is a cryogenic temperature rated epoxy-based material with a dielectric strength of 18 kV/mm. Thus, 0.5-mm thick coating is enough to withstand 9 kV, higher than the required voltage isolation of 8 kV. The selected epoxy material can also survive thermal cycling without forming cracks.

C. Electric Field Simulation

Due to unavoidable sharp edges, the electric field is nonuniform and a higher than expected electric field may be observed. So, the peak electric field needs to be simulated to ensure that the insulation is adequate. A 2-D Ansys Maxwell simulation model is developed for the cold plate-DBC-PCB assembly, as shown in Fig. 14(a), and the electric field is simulated to check the insulation design. The simulation is performed assuming the worst-case condition with ionized air and PCB filled with copper that has a voltage of 8 kV with respect to the cold plate. The simulation result in Fig. 14(b) shows that the highest peak electric field is 16 kV/mm in epoxy and 11.9 kV/mm in ceramic material. Both peak electric fields are within the limits of the respective insulation materials. The whole module also has a thin transparent coating that protects the components from dust, moisture, and condensation.

V. EXPERIMENTAL RESULTS

The performance of the SSCB module design is verified by developing and testing 750-V, 100-A SSCB module prototypes. The SSCB module is designed with enough energy absorption capability so that the 10-kV, 100-A SSCB, consisting of 14 modules, can clear a 1-kA fault current within 100 μ s in the presence of a line inductance of 50 μ H. Two such modules are

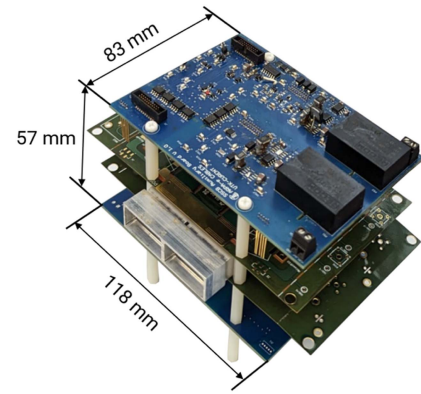


Fig. 15. Developed prototype with two SSCB module and one cold plate.

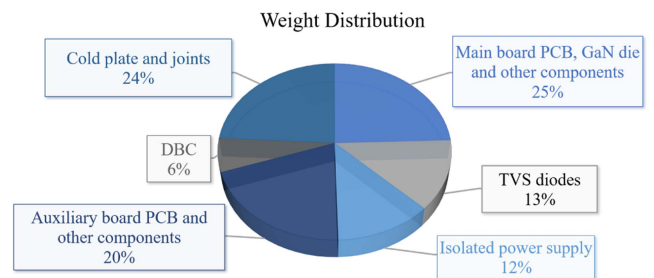


Fig. 16. Weight distribution of the SSCB module.

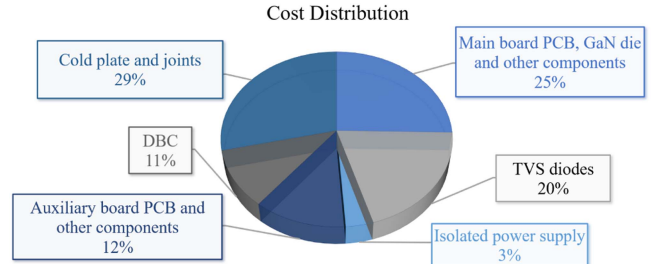


Fig. 17. Cost distribution of the SSCB module.

assembled with a set of cold plates and joints, as shown in Fig. 15. The weight of the two modules with one cold plate assembly is 456 g, leading to a specific power density of 329 kW/kg, higher than any reported specific power density of SSCBs [17]. The distribution of weight among different components of the SSCB module is shown in Fig. 16, which shows that components like isolated power supply and TVS diodes can also have significant weight contributions. The cost distribution of the module is shown in Fig. 17.

A. Turn-Off Test of the Module

Since the SSCB module is designed to carry up to $10\times$ rated current for a short time period to allow any transient or inrush current, avoiding any nuisance trip, the module has to open under a fault current of up to 1 kA. Thus, each of the two paralleled dies must be able to turn OFF 500-A fault current. As this current is ten times higher than the normal operating condition, it is the most

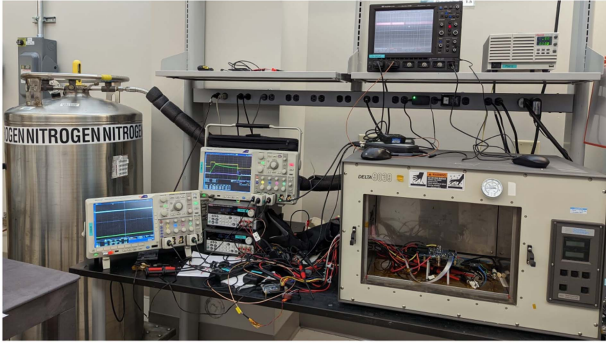


Fig. 18. Test setup for testing an SSCB module in a thermal chamber at -180°C .

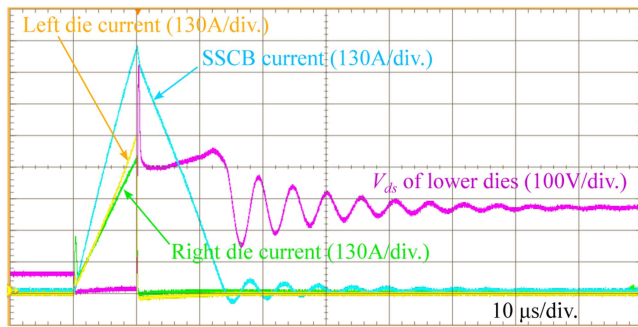


Fig. 19. Voltage and currents during turn-off transition at 940 A fault current at -180°C .

challenging part of the SSCB operation. During this high current turn-OFF transition, the current distribution in the paralleled dies should be reasonably well balanced to avoid damaging one of the dies. To ensure this, the module is first tested in a thermal chamber without DBC and cold plate connected, which allows to measure the individual die currents by placing a Rogowski coil around each die. Then, the module is attached to DBC and cold plate and tested with LN_2 cooling through the cold plate, which is a more realistic operating condition of the SSCB module.

1) *Test in Thermal Chamber:* The prototype is tested up to 400 A at room temperature and 1 kA at cryogenic temperature in a thermal chamber. The developed test setup is shown in Fig. 18. The thermal chamber uses LN_2 to create a low-temperature environment. All electrical connections to the SSCB module go through a slot on the left wall of the chamber. Then, the slot is sealed with fiberglass insulation and the front door of the chamber is closed to create a fully sealed cryogenic test chamber. The performance of the module is evaluated at different temperatures from room temperature down to -180°C , which is the expected die temperature of the LN_2 cooled module.

The test waveforms in Fig. 19 show a turn-OFF transition of the SSCB module interrupting a fault current of 1 kA at -180°C . Since the SSCB has to absorb the energy of a line inductance of $50\text{ }\mu\text{H}$, each of its modules has to absorb the energy stored in $3.6\text{-}\mu\text{H}$ line inductance. Hence, the module is tested here using a $4\text{-}\mu\text{H}$ line inductance and a dc source up to 550 V. However, the additional wire connections in the test setup increase the line

inductance to $5.6\text{ }\mu\text{H}$. At the beginning of this test, the SSCB module GaN devices are turned ON under a short-circuit fault condition. When the current reaches the threshold for SSCB tripping, the GaN devices are turned OFF simultaneously, and die currents fall to zero. The voltage across the GaN devices increases until it is clamped by the breakdown of the TVS diode. It should be noted that the peak voltage across the die is observed when the current transfers from the snubber branch to the TVS branch. During this transition, the GaN device is already fully turned OFF and the voltage spike observed is mainly determined by the characteristics of the TVS diode. The peak voltage observed across the GaN devices under 1-kA current interruption condition is 716 V, which is higher than the voltage rating of the GaN device. However, the datasheet of the selected GaN device states that the device can withstand 750 V for $1\text{ }\mu\text{s}$ [31]. During the turn-OFF transition of the module test, the drain to source voltage of the GaN device falls back below 650 V within 300 ns and no failure or loss of performance has been observed due to this short-duration overvoltage. The TVS diode absorbs the stored energy in the line inductance and the fault current is reduced to zero within $14\text{ }\mu\text{s}$.

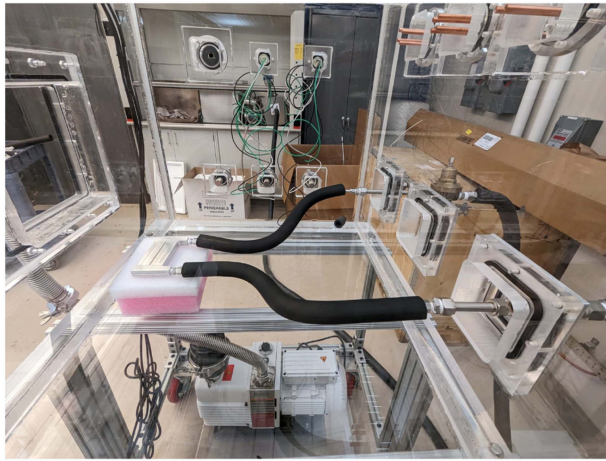
It can be observed that the total current through SSCB is shared well by two parallel GaN dies. The observed imbalance in die currents is below 10% of the SSCB current, which does not cause damage or current saturation of the die at cryogenic temperature. The module is then tested with cold plate cooling, where individual die currents cannot be measured due to cold plate attachment.

2) *Test With Cold Plate:* For this test, a cold-plate-based cooling setup is developed inside a transparent acrylic chamber, as shown in Fig. 20. The cold plates are interfaced to the LN_2 carrying tube using the tube interface shown in Fig. 8. The LN_2 carrying tubes are connected to external LN_2 cylinder and exhaust through the chamber feed-throughs. The tubes are fitted with thermal insulation sleeves. The module is placed on the cold plate and styrofoam-based thermal insulation is added all around the cold plate and module, as shown in Fig. 20(b). The flow of LN_2 is controlled using cryogenic valves at the inlet and outlet of the chamber. Tests are performed when the cold plate temperature reaches a steady state.

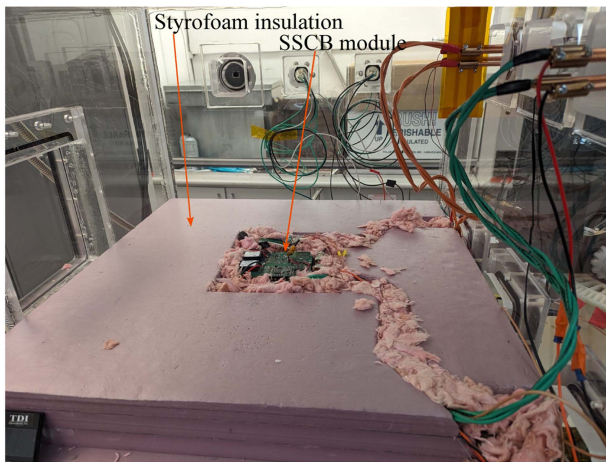
The fault condition is created in the same way as the thermal chamber test and the SSCB interrupts the fault current when it exceeds a preset threshold. The SSCB module is tested at different fault currents to verify its operation with cold plate cooling. The turn-OFF transition waveforms of the module at different fault currents are shown in Fig. 21, which shows that the SSCB module interrupts the fault currents of 525 A, 843 A, and 1 kA in 6, 13, and $18\text{ }\mu\text{s}$, respectively, for a line inductance of $5.5\text{ }\mu\text{H}$. The peak voltage stress on the GaN die increases from 670 V at 525 A to 702 V at 843 A, and 718 V at 1 kA, which is still lower than the short-duration voltage stress limit of the die.

B. Module Insulation Test

The exposed cold plate near the module copper pads is coated with cryogenic epoxy material. Another thin transparent epoxy



(a)



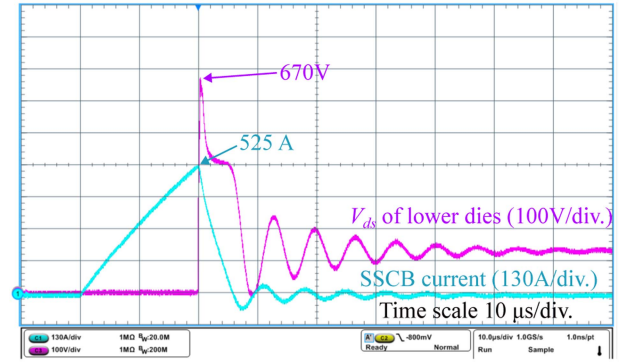
(b)

Fig. 20. Test setup. (a) Cold plate connected to LN_2 cooling system. (b) SSCB module on cold plate with thermal insulation.

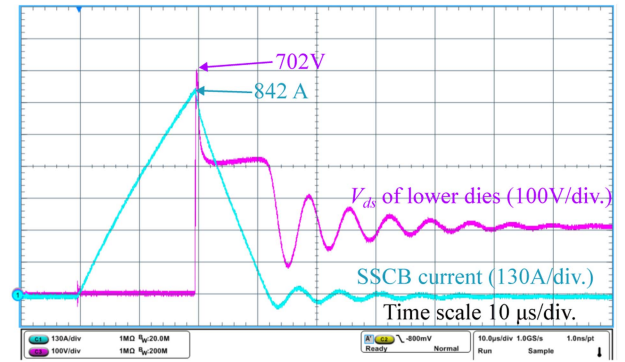
coating is applied on the rest of the SSCB module PCB, as shown in Fig. 22, to protect the module from the effect of dust contamination and condensation at low temperatures. The module needs to withstand 6 kV considering 120% abnormal steady-state voltage. So, the insulation is tested up to 8 kV to ensure some safety margin for reliability. Since the SSCB is designed for a dc distribution system, a dc hipot tester is used to apply a voltage of up to 8 kV on the SSCB PCB with respect to the cold plate. The applied voltage is increased from 1 kV in steps of 1 kV. The leakage current is measured after a few minutes of applying each voltage level to allow the leakage current to reach a steady state. The observed leakage currents during the insulation test are plotted in Fig. 23. It can be observed that the leakage current at 8 kV has a low value of 40 nA.

C. Efficiency Measurement

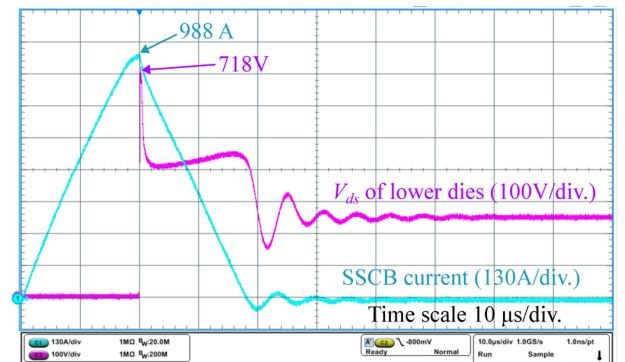
The efficiency of the module is measured with cold plate cooling. The module GaN dies are kept turned ON and a load current is applied by a controllable current source. The load



(a)



(b)



(c)

Fig. 21. Experimental waveforms during turn-off transition of SSCB module at (a) 525 A, (b) 843 A, and (c) 1-kA fault current.

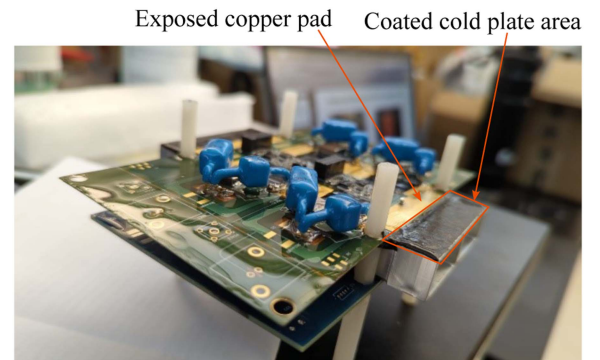


Fig. 22. SSCB module with coating for insulation test.

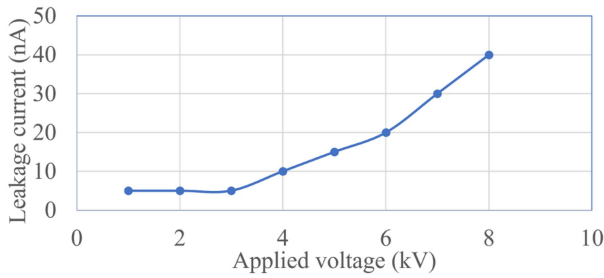


Fig. 23. Observed leakage currents during insulation tests.

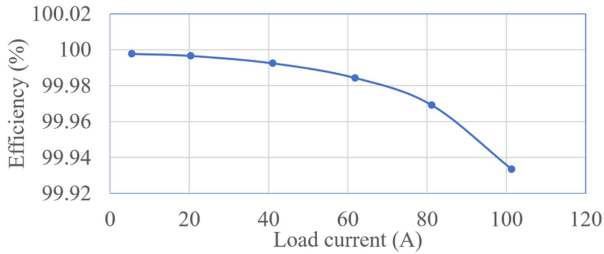


Fig. 24. Measured efficiency of the SSCB module prototype.

current is increased in steps up to the rated current of 100 A. For each current step, the efficiency is measured after the voltage drop reaches its steady value. The measured efficiencies are plotted in Fig. 24. The efficiency at the rated current of 100 A is 99.93%. The measured loss in efficiency evaluation also includes the loss in the auxiliary circuits, which contributes about 3% of the loss at the rated current condition.

This work targets future electric aircraft propulsion, where cryogenic fluids, such as liquid hydrogen or liquified natural gas are expected to be used as fuel. The fuel cell or the engine does not need the fuel to be at cryogenic temperature. Hence, the fuel does not go through a refrigeration process after it is used to cool the SSCB. In this case, there is no additional energy loss due to the cooling system of SSCB. However, in the absence of cryogenic fuel, the SSCB will need closed cooling with a refrigeration system and the additional loss in the cooling process should also be considered [25], [45].

VI. CONCLUSION

The design of a 750-V/100-A cryogenic SSCB module using GaN devices for MVdc applications in EAP systems is presented. The module structure design and compact implementation are discussed aiming at high specific power density. The modular cold plate design for cryogenic cooling and insulation design for 10-kV bipolar MVdc application are discussed. Thermal interface design is proposed to withstand wide thermal cycling in cryogenic applications. DBC and epoxy coating-based insulation is developed to achieve compact design of the module. High specific power density and efficiency are verified with a developed SSCB module prototype that achieves a power density of 329 kW/kg and an efficiency of 99.93%. The 1-kA fault current interruption performance of the designed module is experimentally validated. The insulation design test

up to 8 kV shows a low leakage current. Overall, this work provides compact mechanical, thermal, and insulation design to achieve high specific power density and demonstrates the benefits of cryogenically cooled GaN-based SSCB modules for MVdc application.

ACKNOWLEDGMENT

The authors would also like to thank Dr. John Hull from Boeing Inc. for helping in design and development of the cryogenic cooling system.

REFERENCES

- [1] R. Jansen, C. Bowman, A. Jankovsky, R. Dyson, and J. Felder, "Overview of NASA electrified aircraft propulsion (EAP) research for large subsonic transports," in *Proc. 53rd AIAA/SAE/ASME Joint Propulsion Conf.*, 2017, pp. 1–25.
- [2] D. Izquierdo, A. Barrado, C. Raga, M. Sanz, and A. Lazaro, "Protection devices for aircraft electrical power distribution systems: State of the art," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 47, no. 3, pp. 1538–1550, Jul. 2011.
- [3] T. C. Cano et al., "Future of electrical aircraft energy power systems: An architecture review," *IEEE Trans. Transport. Electrification*, vol. 7, no. 3, pp. 1915–1929, Sep. 2021.
- [4] Z. Zhang, X. Li, M. Chen, Y. Huang, S. Xu, and X. Zhao, "Research on critical technical parameters of HVDC circuit breakers applied in Nan'ao multi-terminal VSC-HVDC project," *Dianwang Jishu/Power Syst. Technol.*, vol. 41, pp. 2417–2422, 2017.
- [5] Z. Yu, X. Yan, X. Zhang, L. Qu, Z. Gan, and Y. Huang, "The design and development of a novel 10 kV/60 kA hybrid DC circuit breaker based on mixed solid-state switches," *IEEE Trans. Ind. Electron.*, vol. 70, no. 3, pp. 2440–2449, Mar. 2023.
- [6] R. Rodrigues, Y. Du, A. Antoniazzi, and P. Cairolis, "A review of solid-state circuit breakers," *IEEE Trans. Power Electron.*, vol. 36, no. 1, pp. 364–377, Jan. 2021.
- [7] A. R. F. Bento, F. Bento, and A. J. M. Cardoso, "A review on hybrid circuit breakers for DC applications," *IEEE Open J. Ind. Electron. Soc.*, vol. 4, pp. 432–450, 2023.
- [8] D. Keshavarzi, T. Ghanbari, and E. Farjah, "A Z-source-based bidirectional DC circuit breaker with fault current limitation and interruption capabilities," *IEEE Trans. Power Electron.*, vol. 32, no. 9, pp. 6813–6822, Sep. 2017.
- [9] X. Diao, F. Liu, Y. Zhuang, S. Pan, and X. Zha, "A bidirectional efficient circuit breaker with automatic and controllable shutoff functions for DC distribution network," *IEEE Trans. Ind. Electron.*, vol. 70, no. 8, pp. 7988–7999, Aug. 2023.
- [10] Z. Zhou, J. Jiang, S. Ye, C. Liu, and D. Zhang, "A γ -source circuit breaker for DC microgrid protection," *IEEE Trans. Ind. Electron.*, vol. 68, no. 3, pp. 2310–2320, Mar. 2021.
- [11] X. Xu, J. Ye, Y. Wang, X. Xu, Z. Lai, and X. Wei, "Design of a reliable bidirectional solid-state circuit breaker for DC microgrids," *IEEE Trans. Power Electron.*, vol. 37, no. 6, pp. 7200–7208, Jun. 2022.
- [12] D. Zhang, Y. Xu, J. Brandt, Z. Zhang, D. Qin, and D. Dong, "A solid-state circuit breaker without current limiting inductor," *IEEE Trans. Ind. Appl.*, vol. 59, no. 4, pp. 4640–4650, Jul./Aug. 2023.
- [13] T. Pang and M. D. Manjrekar, "A surgeless diode-clamped multi-level solid-state circuit breaker for medium-voltage DC distribution systems," *IEEE Trans. Ind. Electron.*, vol. 69, no. 7, pp. 7329–7339, Jul. 2022.
- [14] U. Mehrotra, B. Ballard, and D. C. Hopkins, "Bidirectional solid-state circuit breaker using super cascode for MV SST and energy storage systems," *IEEE Trans. Emerg. Sel. Topics Power Electron.*, vol. 10, no. 4, pp. 3652–3661, Aug. 2022.
- [15] L. A. Garcia Rodriguez et al., "Cascaded SiC JFET topology for high-voltage solid-state circuit breaker applications," *IEEE Trans. Ind. Appl.*, vol. 59, no. 2, pp. 2326–2339, Mar./Apr. 2023.
- [16] G. E. Mejia-Ruiz, M. R. A. Paternina, A. Zamora-Mendez, J. C. Rosas-Caro, and G. Bolivar-Ortiz, "A novel GaN-based solid-state circuit breaker with voltage overshoot suppression," *IEEE Trans. Ind. Electron.*, vol. 69, no. 9, pp. 8949–8960, Sep. 2022.

- [17] Z. Dong, R. Ren, and F. Wang, "Development of high-power bidirectional DC solid-state power controller for aircraft applications," *IEEE Trans. Emerg. Sel. Topics Power Electron.*, vol. 10, no. 5, pp. 5498–5508, Oct. 2022.
- [18] Q. Guo, J. Zhang, and T. Chi, "Review of DC circuit breaker technology," in *Proc. IEEE Sustain. Power Energy Conf.*, 2022, pp. 1–5.
- [19] Y. Wei, L. Du, X. Du, and A. Mantooth, "Multi-level active gate driver for SiC MOSFETs with paralleling operation," in *Proc. IEEE 22nd Workshop Control Modelling Power Electron.*, 2021, pp. 1–7.
- [20] Z. Dong, R. Ren, W. Zhang, F. F. Wang, and L. M. Tolbert, "Instability issue of paralleled dies in a SiC power module in solid-state circuit breaker applications," *IEEE Trans. Power Electron.*, vol. 36, no. 10, pp. 11763–11773, Oct. 2021.
- [21] F. F. Wang, R. Chen, and Z. Dong, "Power electronics: A critical enabler of future hydrogen–electric systems for aviation," *IEEE Electrific. Mag.*, vol. 10, no. 2, pp. 57–68, Jun. 2022.
- [22] H. Gui et al., "Review of power electronics components at cryogenic temperatures," *IEEE Trans. Power Electron.*, vol. 35, no. 5, pp. 5144–5156, May 2020.
- [23] R. Ren et al., "Characterization of 650 V enhancement-mode GaN HEMT at cryogenic temperatures," in *Proc. IEEE Energy Convers. Congr. Expo.*, 2018, pp. 891–897.
- [24] L. Nela, N. Perera, C. Erine, and E. Matioli, "Performance of GaN power devices for cryogenic applications down to 4.2 K," *IEEE Trans. Power Electron.*, vol. 36, no. 7, pp. 7412–7416, Jul. 2021.
- [25] Z. Zheng et al., "An ultralow-loss fault-current-limiting z-source LVDC circuit breaker using cryogenic power electronics," *IEEE Trans. Power Electron.*, vol. 38, no. 5, pp. 5626–5633, May 2023.
- [26] T. Arvin, J. He, and K. Waters, "Solid-state DC circuit breaker based on HTS fault current limiter and SiC MOSFET modules," in *Proc. IEEE Transp. Electrific. Conf. Expo.*, 2022, pp. 1111–1116.
- [27] "Environmental Conditions and Test Procedures for Airborne Equipment RTCA Standard DO-160 G," 2014.
- [28] S. K. Dam et al., "Module development of a high-power-density high-efficiency cryogenic solid state circuit breaker for electrified aircraft propulsion," in *Proc. IEEE Energy Convers. Congr. Expo.*, 2023, pp. 1996–2001, doi: [10.1109/ECCCE53617.2023.10362337](https://doi.org/10.1109/ECCCE53617.2023.10362337).
- [29] The National Academies of Sciences, Engineering, and Medicine, Commercial Aircraft Propulsion and Energy Systems Research: Reducing Global Carbon Emissions. Washington, DC, USA: The National Academies Press, 2016.
- [30] Fuel Cell Technologies Office, DOE, Washington, DC, USA, "Comparison of fuel cell technologies," [Online]. Available: <https://www.energy.gov/eere/fuelcells/articles/comparison-fuel-cell-technologies-fact-sheet>
- [31] Gan Systems, Infineon Technologies AG, Neubiberg, Germany, "650 V enhancement mode GaN transistor, GS-065-150-1-D2 datasheet," [Online]. Available: <https://gansystems.com/gan-transistors/gs-065-150-1-d2/>
- [32] S. K. Dam et al., "Experimental evaluation of cryogenic performances of electronic components for signal isolation in medium voltage power converters," in *Proc. IEEE Appl. Power Electron. Conf. Expo.*, 2023, pp. 3196–3200.
- [33] "An introduction to transient voltage suppressor devices," Microsemi, Aliso Viejo, CA, USA, Appl. Note 102, 2020. [Online]. Available: https://www.microsemi.com/document-portal/doc_view/14604-micronote-102-an-introduction-to-tvs-devices
- [34] M. Mashaba and K. Nixon, "Deducing metal oxide varistor life span from pulse rating curves for surges of different magnitudes," in *Proc. IEEE Int. Conf. High Voltage Eng. Appl.*, 2016, pp. 1–4.
- [35] C. Zhang, H. Xing, P. Li, C. Li, D. Lv, and S. Yang, "An experimental study of the failure mode of ZnO varistors under multiple lightning strokes," *Electronics*, vol. 8, no. 2, 2019, Art. no. 172.
- [36] "Failure mechanisms for transient voltage suppressors," Microsemi, Aliso Viejo, CA, USA, App. Note 135, 2018. [Online]. Available: https://www.microsemi.com/document-portal/doc_view/129581-micronote-135-failure-mechanisms-for-transient-voltage-suppressors
- [37] Littelfuse, Chicago, IL, USA, "AK3 series," [Online]. Available: https://m.littelfuse.com/media/electronics/datasheets/tvs_diodes/littelfuse_tvs_diode_ak3_datasheet.pdf
- [38] C.-H. Yang et al., "A high efficiency 750 V/100 A cryogenically cooled solid state circuit breaker module for aviation applications," in *Proc. IEEE Energy Convers. Congr. Expo.*, 2023, pp. 1745–1751.
- [39] Y. Fan, J. Chuang, D. Shia, and M. Prabhugoud, "Manifold integrated cold plate for single-phase and two-phase liquid cooling," in *Proc. 22nd IEEE Intersociety Conf. Thermal Thermomechanical Phenomena Electron. Syst.*, 2023, pp. 1–5.
- [40] A. Piazza et al., "Considerations and challenges for large area embedded micro-channels with 3D manifold in high heat flux power electronics applications," in *Proc. 19th IEEE Intersociety Conf. Thermal Thermomechanical Phenomena Electron. Syst. (ITherm)*, 2020, pp. 77–82.
- [41] R. Perkins, H. Roder, D. Friend, and C. Nieto de Castro, "The thermal conductivity and heat capacity of fluid nitrogen," *Physica A: Stat. Mechanics Appl.*, vol. 173, no. 3, pp. 332–362, 1991. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/037843719190368M>
- [42] C. Cavallo, "All about 6061 aluminum (properties, strength and uses)," 2023. [Online]. Available: <https://www.thomasnet.com/articles/metals-metal-products/6061-aluminum/>
- [43] Wikipedia, "FR-4," [Online]. Available: <https://en.wikipedia.org/wiki/FR-4>
- [44] Remtec, Norwood, MA, USA, "Direct bond copper substrates: Specifications and design rules," [Online]. Available: <https://remtec.com/wp-content/uploads/2023/04/DBC-Design-Rules-R3.pdf>
- [45] S. Büttner and M. März, "Profitability of low-temperature power electronics and potential applications," *Cryogenics*, vol. 121, 2022, Art. no. 103392.



Shimul K. Dam (Member, IEEE) received the B.E. degree from Jadavpur University, Kolkata, West Bengal, India, in 2013, the M.E. and Ph.D. degrees from the Indian Institute of Science, Bangalore, Karnataka, India, in 2015 and 2020, respectively, all in electrical engineering.



He was a Research Associate with the Indian Institute of Science, from 2020 to 2021. From 2021 to 2023, he was with the University of Tennessee, Knoxville, TN, USA, where he has been a Research Assistant Professor with CURENT, since 2023. His research interests include cryogenic power electronics, packaging and insulation design for medium voltage, ac–ac converters, multilevel converters, and solid-state circuit breakers.



Ching-Hsiang Yang (Student Member, IEEE) received the B.S. and M.S. degrees in electrical engineering from the National Tsing Hua University, Hsinchu, Taiwan, in 2012 and 2014, respectively. He is currently working toward the Ph.D. degree in electrical engineering with the Department of Electrical Engineering and Computer Science, University of Tennessee, Knoxville, TN, USA.



In 2014, he joined the Industrial Automation Department at Delta Electronics, Taoyuan, Taiwan, as a Senior Engineer, focusing on grid-connected converter applications and system design. From 2019 to 2021, he served as a Senior Engineer with the Advanced Technology Center, Chicony Power Technology, New Taipei City, Taiwan, where he specialized in integrated motor drive control and system design. His research interests include wide-bandgap devices and applications, medium voltage power electronics, and cryogenic power electronics.



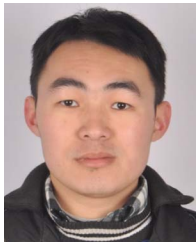
Zhou Dong (Senior Member, IEEE) received the B.S. and M.S. degrees in electrical engineering from the Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2014 and 2017, respectively, and the Ph.D. degree in electrical engineering from the University of Tennessee, Knoxville, TN, USA, in 2022.

He is currently a Senior Scientist with ABB U.S. Corporate Research Center, Raleigh, NC, USA. His research interests include wide bandgap semiconductor device applications, electrical power system protection, and design automation in power electronics.



Dehao Qin (Student Member, IEEE) received the B.S. and M.Sc. degrees in electrical engineering from Northeastern University, Shenyang, China, in 2017 and 2020, respectively, and the Ph.D. degree in electrical engineering from Clemson University, Clemson, SC, USA, in 2023.

His research interests include solid-state circuit breakers, bidirectional dc-dc converters, and SiC-based power electronics.



Ruirui Chen (Member, IEEE) received the B.S. degree from the Huazhong University of Science and Technology, Wuhan, China, and the M.S. degree from Zhejiang University, Hangzhou, China, and the Ph.D. degree from the University of Tennessee, Knoxville, USA, in 2010, 2013, and 2020, respectively, all in electrical engineering.

From 2013 to 2015, he was an Electrical Engineer with FSP-Powerland Technology Inc., Nanjing, China. Since 2020, he has been a Research Assistant Professor with the Department of Electrical Engineering and Computer Science, University of Tennessee, Knoxville, USA. His research interests include wide bandgap devices and applications, medium voltage power electronics, cryogenic power electronics, multilevel converters, and high power high density converters and circuit breakers for electrified transportation and grid applications.



Fred Wang (Fellow IEEE) received the B.S. degree from Xi'an Jiaotong University, Xi'an, China, in 1982, and the M.S. and Ph.D. degrees from the University of Southern California, Los Angeles, CA, USA, in 1985, and 1990, respectively, all in electrical engineering.

He was a Research Scientist with the Electric Power Lab, University of Southern California, from 1990 to 1992. He joined the GE Power Systems Engineering Department, Schenectady, NY, USA, as an Application Engineer in 1992. From 1994 to 2000, he was a Senior Product Development Engineer with GE Industrial Systems, Salem, VA, USA. During 2000 to 2001, he was the Manager of Electronic & Photonic Systems Technology Lab, GE Global Research Center, Schenectady, NY, USA, and Shanghai, China. In 2001, he joined the Center for Power Electronics Systems (CPES), Virginia Tech, Blacksburg, VA, USA, as a Research Associate Professor and became an Associate Professor in 2004. From 2003 to 2009, he also served as the CPES Technical Director. Since 2009, he has been with The University of Tennessee, TN, USA, as a Professor and the Condra Chair of Excellence in Power Electronics. He is a co-founder and the Technical Director of the Center for Ultra-wide-area Resilient Electric Energy Transmission Networks led by The University of Tennessee. His research interests include wide bandgap power electronics, and power electronics applications for transportation and electric grid.

Dr. Wang is a Fellow of the U.S. National Academy of Inventors.

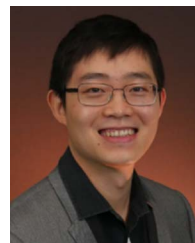


Hua (Kevin) Bai (Senior Member, IEEE) received the B.S. and Ph.D. degrees in electrical engineering from the Department of Electrical Engineering, Tsinghua University, Beijing, China, in 2002 and 2007, respectively.

In 2010, he joined Kettering University (former General Motor Institute) as an Assistant Professor and is currently the Associate Professor with the Department of Electrical Engineering and Computer Science, The University of Tennessee, Knoxville, TN, USA, since 2018. He has authored 2 books, 140 IEEE

papers and hold 20 industrial patents. His research interests include power electronics (particularly wide-bandgap devices) with motor drives, electric vehicle battery chargers, auxiliary power modules and battery energy storage systems.

Dr. Bai is an Associate Editor for the *SAE International Journal of Electrified Vehicles*, and an Associate Editor for IEEE TRANSACTIONS ON TRANSPORTATION ELECTRIFICATION.



Zheyu Zhang (Senior Member, IEEE) received the B.S. and M.S. degrees from the Huazhong University of Science and Technology, Wuhan, China, and the Ph.D. degree from The University of Tennessee, Knoxville, TN, in 2008, 2011, and 2015, respectively, all in electrical engineering.

He was a Research Assistant Professor with the Department of Electrical Engineering and Computer Science, University of Tennessee, Knoxville, TN, USA, from 2015 to 2018; Lead Power Electronics Engineer of General Electric Research with Niskayuna, NY, USA, from 2018 to 2019; and Warren H. Owen' Duke Energy Assistant Professor of Engineering with Clemson University, Clemson, SC, USA, from 2019 to 2023. He is an Assistant Professor with Rensselaer Polytechnic Institute, Troy, NY, US. He has authored or coauthored 100+ papers in the most prestigious journals and conference proceedings, filed 10+ patent applications, authored one book and one book chapter, and presented 10+ IEEE tutorial seminars and webinars. His research interests include wide bandgap-based power electronics characterization and applications.

Dr. Zhang was the recipient of four prize paper awards from the IEEE Industry Applications Society (IAS) and IEEE Power Electronics Society, 2021 IEEE IAS Andrew W. Smith Outstanding Young Member Achievement Award, and 2022 NASA Early Career Faculty Award. He is currently an Associate Editor for IEEE TRANSACTIONS ON POWER ELECTRONICS and IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS. He is currently the Standard Vice-Chair of IEEE IAS Power Electronics Devices and Components Committee.